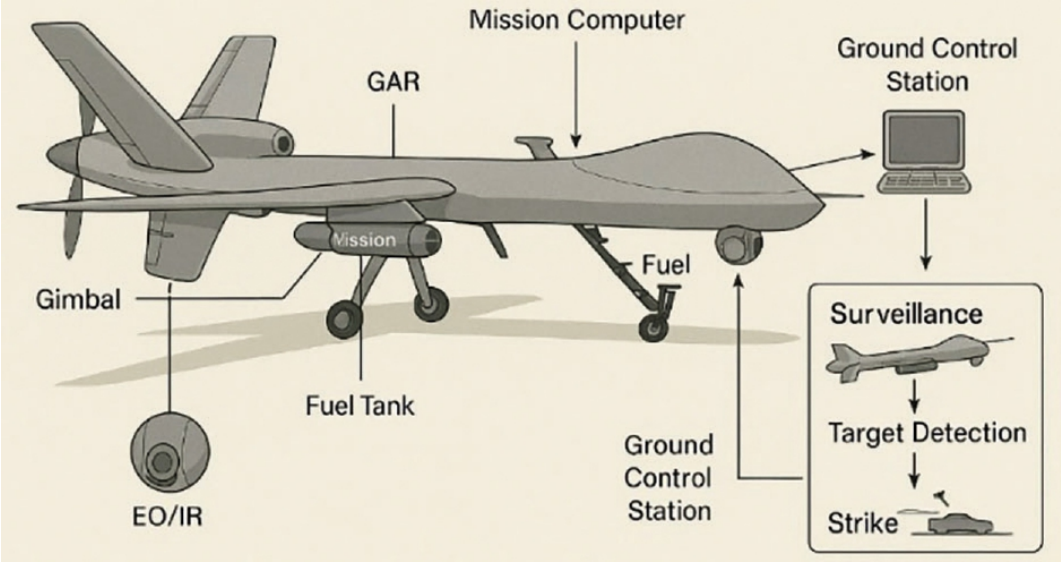



**Figure 1.5** Third-generation drone ISR and combat capabilities. ⏎

### 1.4.4 Fourth generation: Semi-autonomy and commercial applications

The fourth generation of drone technology signifies a critical transition from military-dominated platforms to widespread commercial and civilian adoption, driven by advancements in automation, sensing, and data analytics.

Emerging in the early 2010s, this generation introduced semi-autonomous flight capabilities, where drones could perform pre-programed missions and adaptive maneuvers with minimal human intervention. The integration of on-board processors, real-time data links, and advanced control algorithms allowed drones to execute complex operations with enhanced precision, reliability, and safety.

Technologically, fourth-generation drones are equipped with multi-sensor fusion systems including GPS, LiDAR, ultrasonic sensors, cameras, and inertial measurement units (IMUs) that enable obstacle detection, terrain following, and autonomous navigation in diverse environments. These systems are managed through flight controllers and AI-assisted software